



## **KAPE**

An introduction to Kroll Artifact Parser and Extractor (KAPE) for collecting and processing forensic artifacts

## **Task 1 – Learning Objectives**

### **Revisiting Windows Forensics**

In the [Windows Forensics 1](#) and [Windows Forensics 2](#) rooms, we learned about the different artifacts which store information about a user's activity on a system. We also learned where those artifacts are located and how they can be accessed and interpreted. However, we did all that manually. In many cases, a forensic investigator doesn't have the luxury to perform manual analysis, which can be time-consuming. In such scenarios, it is helpful to have some tools help in automating the evidence collection, sanitization, and processing part.

### **Learning Objectives:**

In this room, we will:

- Learn about KAPE
- How KAPE works
- The different targets and modules used by KAPE
- Collection and analysis of forensic data using KAPE

## **Answer the questions below**

I have already completed the Windows Forensics 1 and Windows Forensics 2 rooms.

## Task 2 – Introduction to KAPE

### Introduction to KAPE:

Kroll Artifact Parser and Extractor (KAPE) parses and extracts Windows forensics artifacts. It is a tool that can significantly reduce the time needed to respond to an incident by providing forensic artifacts from a live system or a storage device much earlier than the imaging process completes.

KAPE serves two primary purposes, 1) collect files and 2) process the collected files as per the provided options. For achieving these purposes, KAPE uses the concept of targets and modules. Targets can be defined as the forensic artifacts that need to be collected. Modules are programs that process the collected artifacts and extract information from them. We will learn about them in the upcoming tasks.

### How it works

KAPE is extensible and highly configurable. In essence, the KAPE binary collects files and processes them as per the provided configuration.

The collection of files (targets) KAPE adds the files to a queue and copies them in two passes. In the first pass, it copies the files that it can. This works for files that the OS has not locked. The rest of the files are passed into a secondary queue. The secondary queue is processed using a different technique that uses raw disk reads to bypass the OS locks and copy the files. The copied files are saved with original timestamps and metadata and stored in a similar directory structure.

Once the data is collected, KAPE can process it using modules. The modules can be independent binaries that run on the collected data and process them to extract information. For example, KAPE will collect and copy the Prefetch file to our target destination during the target collection. Running a Prefetch Parser (PECmd) module on this target will extract the prefetch file and save it in a CSV file.



As the above image shows, KAPE can extract targets from a Live system, a mounted image, or the [F-response](#) utility. KAPE does not need to be installed. It is portable and can be used from network locations or USB drives. To proceed further, click the Start Lab Machine button on the top-right corner to start the attached VM in split-screen mode. Alternatively, you can log in to the machine using the following credentials:

**Username:** \*\*\*\*

**Password:** \*\*\*\*

In the attached VM, you will find KAPE on the Desktop in the folder titled KAPE. In this folder, you will find the following files and directories:

| Name               | Date modified     | Type               | Size      |
|--------------------|-------------------|--------------------|-----------|
| Documentation      | 11/4/2021 4:24 PM | File folder        |           |
| Modules            | 11/4/2021 4:24 PM | File folder        |           |
| Targets            | 11/4/2021 4:24 PM | File folder        |           |
| ChangeLog.txt      | 11/4/2021 4:24 PM | Text Document      | 21 KB     |
| Get-KAPEUpdate.ps1 | 11/4/2021 4:24 PM | Windows PowerSh... | 17 KB     |
| gkape.exe          | 11/4/2021 4:24 PM | Application        | 61,588 KB |
| gkape.settings     | 11/4/2021 6:42 PM | SETTINGS File      | 1 KB      |
| kape.exe           | 11/4/2021 4:24 PM | Application        | 7,006 KB  |

You can see two binaries in this directory, kape.exe and gkape.exe. The first is the CLI version of KAPE, and the second is a GUI version (symbolized by the 'g' prefix).

gkape.settings stores the default settings of the GUI version.

Get-KAPEUpdate.ps1, as the name suggests, is a PowerShell script that checks and downloads updates.

ChangeLog.txt and Documentation are self-explanatory. We will explore Targets and Modules in the following tasks.

## Answer the questions below

From amongst kape.exe and gkape.exe, which binary is used to run GUI version of KAPE?

```
PS C:\Users\THM-4n6\Desktop\KAPE> ls

Directory: C:\Users\THM-4n6\Desktop\KAPE

Mode                LastWriteTime         Length Name
----                -
d-----          2/15/2022  10:55 AM             Documentation
d-----          2/15/2022  12:00 PM             Modules
d-----          2/15/2022  12:15 PM             Targets
-a----          11/16/2021   4:13 PM           21008 ChangeLog.txt
-a----          10/21/2021  10:21 AM           16537 Get-KAPEUpdate.ps1
-a----          11/16/2021   4:10 PM       63103792 gkape.exe
-a----           2/15/2022   9:19 AM           543 gkape.settings
-a----          11/16/2021   4:10 PM       7174968 kape.exe
```

The GUI version of KAPE is started using **gkape.exe**. In the KAPE folder, both **kape.exe** and **gkape.exe** are present, but **kape.exe** is used for command-line execution, while **gkape.exe** launches the graphical interface.

Answer: gkape.exe

## Task 3 – Target Options

In KAPE's lexicon, Targets are the artifacts that need to be collected from a system or image and copied to our provided destination. For example, as we learned in the last room, Windows Prefetch is a forensic artifact for evidence of execution so that we can create a Target for it. Similarly, we can also create Targets for the registry hives. In short, Targets copy files from one place to another.

When we open the Targets directory of KAPE, this is what we will see:

| Name                            | Date modified     | Type          | Size |
|---------------------------------|-------------------|---------------|------|
| !Disabled                       | 11/4/2021 4:24 PM | File folder   |      |
| Antivirus                       | 11/4/2021 4:24 PM | File folder   |      |
| Apps                            | 11/4/2021 4:24 PM | File folder   |      |
| Browsers                        | 11/4/2021 4:24 PM | File folder   |      |
| Compound                        | 11/4/2021 4:24 PM | File folder   |      |
| Logs                            | 11/4/2021 4:24 PM | File folder   |      |
| P2P                             | 11/4/2021 4:24 PM | File folder   |      |
| Windows                         | 11/4/2021 4:24 PM | File folder   |      |
| CompoundTargetGuide.guide       | 11/4/2021 4:24 PM | GUIDE File    | 3 KB |
| CompoundTargetTemplate.template | 11/4/2021 4:24 PM | TEMPLATE File | 1 KB |
| TargetGuide.guide               | 11/4/2021 4:24 PM | GUIDE File    | 4 KB |
| TargetTemplate.template         | 11/4/2021 4:24 PM | TEMPLATE File | 1 KB |

The last four files at the bottom are guides and templates to create Targets and Compound Targets of our own. We will discuss Compound Targets later in this task. As you can see, the targets are grouped into different directories. Let's check out the Windows directory to see what we have:

| Name                       | Date modified     | Type        | Size |
|----------------------------|-------------------|-------------|------|
| WSA                        | 11/4/2021 4:24 PM | File folder |      |
| WSL                        | 11/4/2021 4:24 PM | File folder |      |
| \$Boot.tkape               | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| \$J.tkape                  | 11/4/2021 4:24 PM | TKAPE File  | 2 KB |
| \$LogFile.tkape            | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| \$MFT.tkape                | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| \$MFTMirr.tkape            | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| \$SDS.tkape                | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| \$T.tkape                  | 11/4/2021 4:24 PM | TKAPE File  | 2 KB |
| Amcache.tkape              | 11/4/2021 4:24 PM | TKAPE File  | 2 KB |
| ApplicationEvents.tkape    | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| BCD.tkape                  | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| BITS.tkape                 | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| CertUtil.tkape             | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| EncapsulationLogging.tkape | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| EventLogs.tkape            | 11/4/2021 4:24 PM | TKAPE File  | 1 KB |
| EventLogs-RDP.tkape        | 11/4/2021 4:24 PM | TKAPE File  | 2 KB |
| EventTraceLogs.tkape       | 11/4/2021 4:24 PM | TKAPE File  | 2 KB |
| EventTranscriptDB.tkape    | 11/4/2021 4:24 PM | TKAPE File  | 5 KB |

We can see different .tkape extension files. This is how a Target is defined for KAPE. A TKAPE file contains information about the artifact that we want to collect, such as the path, category, and file masks to collect. As an example, below is how the Prefetch Target is defined.

```

Description: Prefetch files
Author: Eric Zimmerman
Version: 1.0
Id: f6715d3f-b8ca-4cc2-9e5e-4ed18e88abbe
RecreateDirectories: true
Targets:
-
  Name: Prefetch
  Category: Prefetch
  Path: C:\Windows\prefetch\
  FileMask: '*.pf'
-
  Name: Prefetch
  Category: Prefetch
  Path: C:\Windows.old\Windows\prefetch\
  FileMask: '*.pf'
# Documentation
# https://forensicswiki.xyz/wiki/index.php?title=Prefetch
# https://www.youtube.com/watch?v=f4RArR\_3zcs
#
# https://nasbench.medium.com/windows-forensics-analysis-windows-artifacts-part-i-i-71b8fa68d8a1
# https://www.sans.org/blog/device-profiling-with-windows-prefetch
# https://www.youtube.com/watch?v=prEghfj3bPI
#
# https://www.sans.org/security-resources/posters/windows-forensic-analysis/170/download

```

This TKAPE file tells KAPE to collect files with the file mask \*.pf from the path C:\Windows\prefetch and C:\Windows.old\prefetch.

Notice that we have the C:\Windows.old path listed here as well. This path contains files retained after Windows has updated to a new version. For forensic analysis, we can also find interesting historical artifacts from this directory.

### Compound Targets:

KAPE also supports Compound Targets. These are Targets that are compounds of multiple other targets. As mentioned in the previous tasks, KAPE is often used for quick triage collection and analysis. The purpose of KAPE will not be fulfilled if we have to collect each artifact individually. Therefore, Compound Targets help us collect multiple targets by giving a single command. Examples of Compound Targets include !BasicCollection, !SANS\_triage and KAPEtrriage. We can view the Compound Targets on the path KAPE\Targets\Compound. The following image shows what a Compound Target for evidence of execution looks like:

```
Description: Evidence of execution related files
```

```
Author: Eric Zimmerman
```

```
Version: 1.0
```

```
Id: 13bale33-4899-4843-adf0-c7e6a20d758a
```

```
RecreateDirectories: true
```

```
Targets:
```

```
-
```

```
    Name: Prefetch  
    Category: Prefetch  
    Path: Prefetch.tkape
```

```
-
```

```
    Name: RecentFileCache  
    Category: ApplicationCompatability  
    Path: RecentFileCache.tkape
```

```
-
```

```
    Name: Amcache  
    Category: ApplicationCompatability  
    Path: Amcache.tkape
```

```
-
```

```
    Name: Syscache  
    Category: Syscache  
    Path: Syscache.tkape
```

```
# Documentation
```

```
# N/A
```

The above Compound Target will collect evidence of execution from Prefetch, RecentFileCache, AmCache, and Syscache Targets.

### **!Disabled**

This directory contains Targets that you want to keep in the KAPE instance, but you don't want them to appear in the active Targets list.

### **!Local**

If you have created some Targets that you don't want to sync with the KAPE Github repository, you can place them in this directory. These can be Targets that are specific to your environment. Similarly, anything not present in the Github repository when we update KAPE will be moved to the !Local directory.

## Answer the questions below

What is the file extension for KAPE Targets?

```
PS C:\Users\THM-4n6\Desktop\KAPE\Targets> ls .\Windows\
```

Directory: C:\Users\THM-4n6\Desktop\KAPE\Targets\Windows

| Mode   | LastWriteTime       | Length | Name                    |
|--------|---------------------|--------|-------------------------|
| d----  | 2/15/2022 12:19 PM  |        | WSA                     |
| d----  | 2/15/2022 12:19 PM  |        | WSL                     |
| -a---- | 11/12/2021 12:29 PM | 341    | \$Boot.tkape            |
| -a---- | 11/12/2021 12:29 PM | 1947   | \$J.tkape               |
| -a---- | 11/12/2021 12:29 PM | 624    | \$LogFile.tkape         |
| -a---- | 11/12/2021 12:29 PM | 836    | \$MFT.tkape             |
| -a---- | 11/12/2021 12:29 PM | 843    | \$MFTMirr.tkape         |
| -a---- | 11/12/2021 12:29 PM | 891    | \$SDS.tkape             |
| -a---- | 11/12/2021 12:29 PM | 1266   | \$T.tkape               |
| -a---- | 11/12/2021 12:29 PM | 1111   | Amcache.tkape           |
| -a---- | 11/12/2021 12:29 PM | 939    | ApplicationEvents.tkape |
| -a---- | 11/12/2021 12:29 PM | 450    | BCD.tkape               |

KAPE Target files use the **.tkape** file extension. These files define what forensic artifacts KAPE should collect from the source system.

Answer: .tkape

What type of Target will we use if we want to collect multiple artifacts with a single command?

```
PS C:\Users\THM-4n6\Desktop\KAPE\Targets> Get-Content .\Compound\CombinedLogs.tkape
Description: Collect Event logs, Trace logs, Windows Firewall and PowerShell console
Author: Mike Cary, Mark Hallman added the USBDeviceLogs target
Version: 1.1
Id: d4fdd600-15b1-4b78-bc77-88e724861d8d
RecreateDirectories: true
Targets:
-
  Name: Windows Event Logs
  Category: EventLogs
  Path: EventLogs.tkape
-
  Name: Event Trace Logs
  Category: EventTraceLogs
  Path: EventTraceLogs.tkape
-
  Name: PowerShell Console Log
  Category: PowerShellConsoleLog
  Path: PowerShellConsole.tkape
-
  Name: Windows Firewall Log
  Category: WindowsFirewallLogs
  Path: WindowsFirewall.tkape
-
  Name: USBDevicesLogs
  Category: USB
  Path: USBDevicesLogs.tkape
# Documentation
# v1.1 - Added the USBDeviceLogs target
```

A Compound Target is used when we want KAPE to collect multiple artifacts with one command. In this example, the file CombinedLogs.tkape includes several targets inside it, such as Windows Event Logs, Event Trace Logs, PowerShell Console Log, Windows Firewall Log, and USBDevicesLogs.

Command used:

```
Get-Content .\Compound\CombinedLogs.tkape
```

Answer: Compound Targets

## Task 4 – Module Options

Modules, in KAPE's lexicon, run specific tools against the provided set of files. Their goal is not to copy files from one place to another but rather run some command and store the output. Generally, the output is in the form of CSV or TXT files.

This is what the Modules directory looks like in KAPE:

| Name                            | Date modified     | Type          | Size |
|---------------------------------|-------------------|---------------|------|
| !Disabled                       | 11/4/2021 4:24 PM | File folder   |      |
| !Local                          | 11/4/2021 4:24 PM | File folder   |      |
| Apps                            | 11/4/2021 4:24 PM | File folder   |      |
| bin                             | 11/4/2021 4:24 PM | File folder   |      |
| Compound                        | 11/4/2021 4:24 PM | File folder   |      |
| EZTools                         | 11/4/2021 4:24 PM | File folder   |      |
| KapeResearch                    | 11/4/2021 4:24 PM | File folder   |      |
| KapeSync                        | 11/4/2021 4:24 PM | File folder   |      |
| Windows                         | 11/4/2021 4:24 PM | File folder   |      |
| CompoundModuleGuide.guide       | 11/4/2021 4:24 PM | GUIDE File    | 2 KB |
| CompoundModuleTemplate.template | 11/4/2021 4:24 PM | TEMPLATE File | 1 KB |
| ModuleGuide.guide               | 11/4/2021 4:24 PM | GUIDE File    | 3 KB |
| ModuleTemplate.template         | 11/4/2021 4:24 PM | TEMPLATE File | 1 KB |

Similar to the previous task, we see guides and templates for creating Modules and Compound Modules. We also see the !Disabled, !Local and Compound directories, which are similar to what we saw in the previous task. We will not discuss these again, as we discussed them in the last task. We see that most of the Modules are grouped together in different directories. One thing we find different is the bin directory. We will discuss that in a bit. For now, let's open the Windows directory and see what we have there:

| Name                                                                                                                        | Date modified     | Type       | Size |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------|------------|------|
|  PowerShell_NetUserAdministrators.mkape    | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  PowerShell_ProcessList_CimInstance.mka... | 11/4/2021 4:24 PM | MKape File | 2 KB |
|  PowerShell_ProcessList_WMI.mkape          | 11/4/2021 4:24 PM | MKape File | 3 KB |
|  PowerShell_WMIRepositoryAuditing.mka...   | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_ARPCache.mkape                    | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_DNSCache.mkape                    | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_IPConfig.mkape                    | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_ManageBDE_BitLockerKeys.mk...     | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_ManageBDE_BitLockerStatus.m...    | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_nbtstat_NetBIOSCache.mkape        | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_nbtstat_NetBIOSSessions.mkape     | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_Net_Accounts.mkape               | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_Net_File.mkape                  | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_Net_LocalGroup.mkape            | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_Net_Session.mkape               | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_Net_Share.mkape                 | 11/4/2021 4:24 PM | MKape File | 1 KB |
|  Windows_Net_Start.mkape                 | 11/4/2021 4:24 PM | MKape File | 1 KB |

Here we see files with the .mkape extension. These are understood as Modules by KAPE. Let's open an MKAPE file and see how it is structured. The following image shows the Windows\_IPConfig MKAPE file.

```
Description: IPConfig
Category: LiveResponse
Author: Mike Cary
Version: 1.0
Id: 5afdfd6b-5ebb-4545-9980-88e6f421e508
ExportFormat: txt
Processors:
    -
      Executable: C:\Windows\System32\ipconfig.exe
      CommandLine: /all
      ExportFormat: txt
      ExportFile: ipconfig.txt
# Documentation
#
https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/ipconfig
```

Notice that the MKAPE file tells KAPE about the executable that has to be run, the command line parameters of the executable file, the output export format, and the filename to export to. But what if the executable that we want to run is not present on the system? This brings us to the bin directory.

### **The bin directory:**

The bin directory contains executables that we want to run on the system but are not natively present on most systems. KAPE will run executables either from the bin directory or the complete path. An example of files to be kept in the bin directory are Eric Zimmerman's tools, which are generally not present on a Windows system. We used them extensively in the Windows Forensics rooms.

| Name                      | Date modified      | Type        | Size     |
|---------------------------|--------------------|-------------|----------|
| EvtxECmd                  | 11/4/2021 4:24 PM  | File folder |          |
| RECmdd                    | 11/4/2021 4:24 PM  | File folder |          |
| SQLCmd                    | 11/4/2021 4:24 PM  | File folder |          |
| AmcacheParser.exe         | 9/21/2021 8:56 AM  | Application | 5,021 KB |
| AppCompatCacheParser.exe  | 9/21/2021 8:58 AM  | Application | 4,894 KB |
| bstrings.exe              | 8/8/2021 6:34 PM   | Application | 2,618 KB |
| JLECmd.exe                | 10/8/2021 12:19 PM | Application | 4,342 KB |
| LECmd.exe                 | 10/8/2021 12:48 PM | Application | 3,357 KB |
| MFTECmd.exe               | 10/7/2021 7:43 PM  | Application | 4,162 KB |
| PECmd.exe                 | 3/20/2021 10:58 AM | Application | 3,965 KB |
| RBCmd.exe                 | 3/20/2021 11:00 AM | Application | 2,676 KB |
| RecentFileCacheParser.exe | 3/20/2021 11:13 AM | Application | 2,216 KB |
| SBECmd.exe                | 4/17/2020 10:47 AM | Application | 4,563 KB |
| SrumECmd.exe              | 10/8/2021 12:00 PM | Application | 4,620 KB |
| SumECmd.exe               | 10/8/2021 12:04 PM | Application | 3,667 KB |
| WxTCmd.exe                | 5/27/2021 8:10 AM  | Application | 5,565 KB |

Notice that most of the binaries present here are from Eric Zimmerman's Tools.

## Answer the questions below

What is the file extension of the Modules files?

```
PS C:\Users\THM-4n6\Desktop\KAPE> ls .\Modules\Windows\

Directory: C:\Users\THM-4n6\Desktop\KAPE\Modules\Windows

Mode                LastWriteTime         Length Name
----                -
-a----           11/12/2021  12:29 PM             607 PowerShell_NetUserAdministrators.mkape
-a----           11/12/2021  12:29 PM          1198 PowerShell_ProcessList_CimInstance.mkape
```

KAPE Module files use the **.mkape** file extension. These files define how KAPE should process collected artifacts after collection. In the screenshot, the files inside the **Modules\Windows** folder both end with **.mkape**.

Answer: mkape

What is the name of the directory where binary files are stored, which may not be present on a typical system, but are required for a particular KAPE Module?

```
PS C:\Users\THM-4n6\Desktop\KAPE> ls .\Modules\bin\

Directory: C:\Users\THM-4n6\Desktop\KAPE\Modules\bin

Mode                LastWriteTime         Length Name
----                -
d-----         2/15/2022  11:12 AM             EvtxECmd
d-----         2/15/2022  11:41 AM             RECcmd
d-----         2/15/2022  11:46 AM             SQLECmd
-a----         9/21/2021   8:56 AM        5140728 AmcacheParser.exe
-a----         9/21/2021   8:58 AM        5010680 AppCompatCacheParser.exe
```

The required executable files for some KAPE Modules are stored in the **Modules\bin** directory. These binaries may not normally exist on a standard Windows system, but KAPE uses them when running specific modules to process collected artifacts.

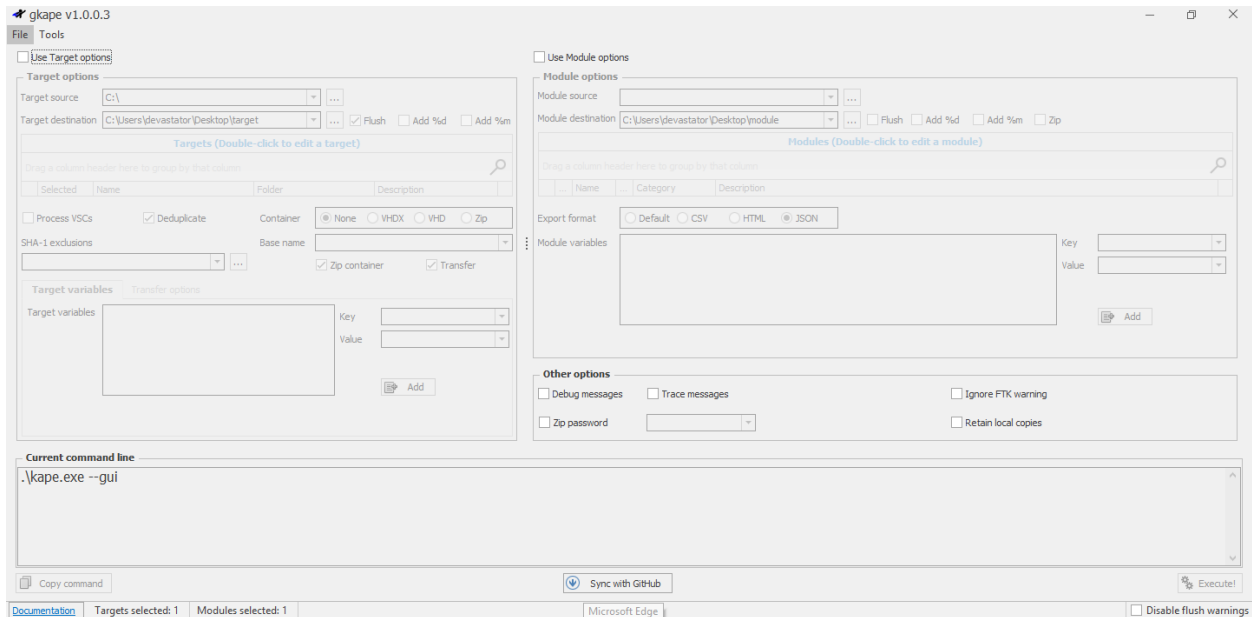
Command used:

```
ls .\Modules\bin\
```

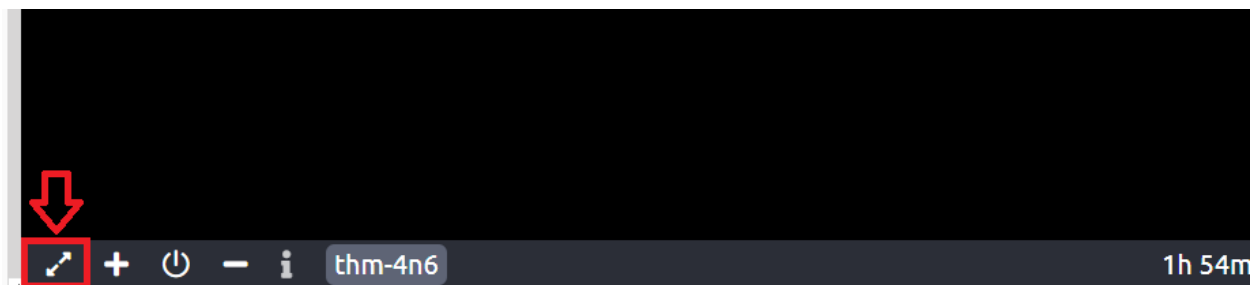
Answer: bin

## Task 5 – KAPE GUI

Now that we have learned about the different components of KAPE let's take it for a test drive. In the attached VM, double-click to open the gkape.exe file. You will see the following Window:



**Tip:** If the Window is not correctly visible in the split-screen, you can open it in a full browser tab by clicking this button. Also try maximizing the gkape.exe window or remove maximizing to be able to view all options.



Here you can see that there are different options, but most are disabled. To collect Targets We will go ahead by enabling the Use Target Options checkbox. This will enable the options present in the left half of the Window:

If we want to perform forensics on the same machine on which KAPE is running, we will provide C:\ for the Target source. We can select the target destination of our choice. All the triage files will be copied to the Target destination that we provide.

**Target options**

Target source:  ...

Target destination:  ... ☒ Flush ☐ Add %d ☐ Add %m

**Targets (Double-click to edit a target)**

Drag a column header here to group by that column 🔍

|   | Selected                            | Name             | Folder   | Description            |
|---|-------------------------------------|------------------|----------|------------------------|
| 🔍 | <input checked="" type="checkbox"/> | REC              | REC      | REC                    |
| ▶ | <input type="checkbox"/>            | !BasicCollection | Compound | Basic Collection       |
|   | <input type="checkbox"/>            | !SANS_Triage     | Compound | SANS Triage Collection |
|   | <input type="checkbox"/>            | \$Boot           | Windows  | \$Boot                 |
|   | <input type="checkbox"/>            | \$J              | Windows  | \$J                    |

Here, the Flush checkbox will delete all the contents of the Target destination, so we have to be careful when using that. We have disabled the Flush checkbox so that it does not delete data already present in the directories. Add %d will append date info to the directory name where the collected data is saved. Similarly, Add %m will append machine info to the Target destination directory. We can select our desired Target from the list shown above. The Search bar helps us search for the names of the desired Targets quickly.

**Targets (Double-click to edit a target)**

Drag a column header here to group by that column

|   | Selected                            | Name       | Folder   | Description                     |
|---|-------------------------------------|------------|----------|---------------------------------|
| 🔍 | <input checked="" type="checkbox"/> | REC        | REC      | REC                             |
| I | <input checked="" type="checkbox"/> | KapeTriage | Compound | Kape Triage collections tha...  |
|   | <input type="checkbox"/>            | Signal     | Apps     | Signal (Please view this tka... |

We can select if we want to process Volume Shadow Copies by enabling Process VSCs. We can select the transfer checkbox if we want to transfer the collected artifacts through an SFTP server or an S3 bucket. For transfer, the files must be enclosed in a container, which can be Zip, VHD, or VHDX. Similarly, we can provide exclusions based on SHA-1, and KAPE will not copy the excluded files. When enclosing in a container, we will need to give a Base name that will be used for all the created files. It is not required if we are not transferring files or enclosing them in a container.

☐ Process VSCs
 ☒ Deduplicate
 Container: ☐ None ☐ VHDX ☐ VHD ☒ Zip

SHA-1 exclusions: 
 Base name:

☒ Zip container
 ☐ Transfer

In the Current command line tab, we can see the command line options being added or removed while configuring the UI. This Window will show more options in the command line as we add options. Please note that the destination path in your case will be different from the one shown in the image. Notice the --tflush flag here. It means that when this command line was created, the Flush checkbox was still checked.

**Current command line**

```
.\kape.exe --tsource C: --tdest C:\Users\Umar\Desktop\kape --tflush --target KapeTriage --gui
```

By checking the Use Module Options checkbox, the right side of the KAPE Window will also be enabled.

gkape v1.1.0.1

File Tools

☒ Use Target options

**Target options**

Target source:

Target destination:

☒ Flush ☐ Add %id ☐ Add %m

**Targets (Double-click to edit a target)**

Drag a column header here to group by that column

| Selected                            | Name       | Folder   | Description                    |
|-------------------------------------|------------|----------|--------------------------------|
| <input checked="" type="checkbox"/> | KapeTriage | Compound | KapeTriage collections that... |
| <input type="checkbox"/>            | Signal     | Apps     | Signal (Please view this...)   |

☐ Process VSCs ☒ Deduplicate Container: ☐ None ☐ VHDX ☐ VHD ☒ Zip

SHA-1 exclusions:  Base name:

☒ Zip container ☐ Transfer

**Target variables**

Transfer options

Target variables:

Key:  Value:

☒ Use Module options

**Module options**

Module source:

Module destination:

☒ Flush ☐ Add %id ☐ Add %m ☐ Zip

**Modules (Double-click to edit a module)**

Drag a column header here to group by that column

| Selected                 | Name      | Folder   | Category         | Description                                                          |
|--------------------------|-----------|----------|------------------|----------------------------------------------------------------------|
| <input type="checkbox"/> | ITTool... | Compound | Sync             | Sync for new Maps, Batch Files, Targets and Modules                  |
| <input type="checkbox"/> | IEZPh...  | Compound | Modules          | Eric Zimmerman Parsers                                               |
| <input type="checkbox"/> | Amcac...  | EZTools  | ProgramExecution | AmcacheParser: extract program execution information                 |
| <input type="checkbox"/> | AppCo...  | EZTools  | ProgramExecution | AppCompatCacheParser: extract AppCompatCache (shimcache) information |

Export format: ☒ Default ☐ CSV ☐ HTML ☐ JSON

Module variables:

Key:  Value:

**Other options**

☐ Debug messages ☐ Trace messages ☐ Ignore FTK warning

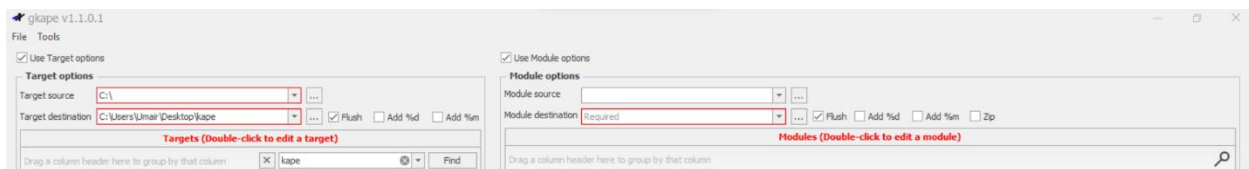
☐ Zip password:  ☐ Retain local copies

**Current command line**

```
.\kape.exe --tsource C: --tdest C:\Users\Umar\Desktop\kape --tflush --target KapeTriage --mflush --gui
```

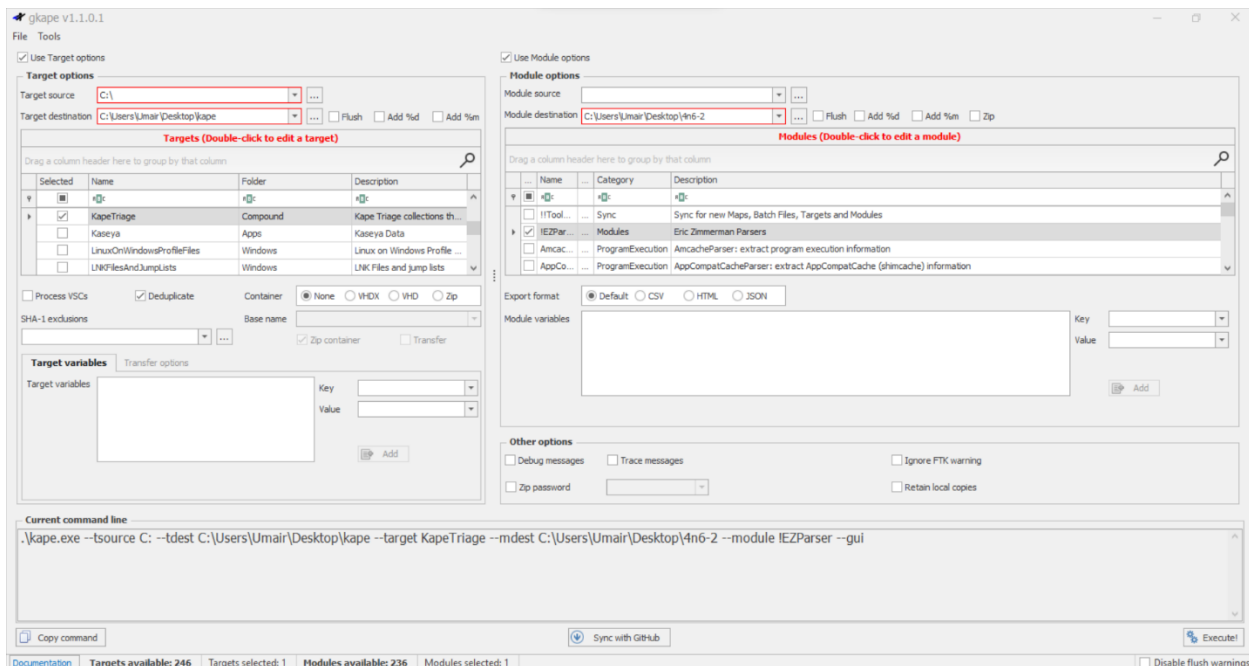
Documentation Targets available: 246 Targets selected: 1 Modules available: 236 Modules selected: 0 ☐ Disable flush warnings

When using both Target and Module Options, providing Module Source is not required. The selected Modules will use the Target destination as the source.



The rest of the options for Modules are similar to the ones for Targets, so we won't go into details for them.

Below you will see what the configuration looks like when we have KAPE all set up for collecting Targets and processing them using Modules.



We have selected the KapeTriage compound Target and !EZParser Compound Module. The command line below shows the CLI command that will be run. The Execute! button in the bottom right corner will execute the command. The Disable flush warnings checkbox underneath it will not warn us when we are using the Flush flags. When we press Execute! We will see a command line window open and show us the logs as KAPE performs its tasks. It will take a few minutes to execute since it will be collecting all the data and then running the module processes on it. Once it completes, it will show us the total execution time, and we can press any key to terminate the command window.

```
D:\Kape\kape.exe

KAPE version 1.1.0.1 Author: Eric Zimmerman (kape@kroll.com)

KAPE directory: D:\KAPE
Command line: --tsource C: --tdest C:\Users\Umais\Desktop\kape --target KapeTriage --mdest C:\Users\Umais\Desktop\4n6-2 --module !EZParser

System info: Machine name: UMAIR-THINKBOOK, 64-bit: True, User: Umais OS: Windows10 (10.0.22000)

Using Target operations
Found 14 targets. Expanding targets to file list...
Target 'ApplicationEvents' with Id '2da16dbf-ea47-448e-a00f-fc442c3109ba' already processed. Skipping!
Target 'ApplicationEvents' with Id '2da16dbf-ea47-448e-a00f-fc442c3109ba' already processed. Skipping!
Target 'ApplicationEvents' with Id '2da16dbf-ea47-448e-a00f-fc442c3109ba' already processed. Skipping!
Target 'ApplicationEvents' with Id '2da16dbf-ea47-448e-a00f-fc442c3109ba' already processed. Skipping!
Target 'ApplicationEvents' with Id '2da16dbf-ea47-448e-a00f-fc442c3109ba' already processed. Skipping!
Found 3,059 files in 4.257 seconds. Beginning copy...
Deferring 'C:\Windows\System32\winevt\logs\Application.evtx' due to IOException...
Deferring 'C:\Windows\System32\winevt\Logs\Microsoft-Windows-Windows Defender%40operational.evtx' due to IOException...
Deferring 'C:\Windows\System32\winevt\Logs\Microsoft-Windows-Windows Defender%4WHC.evtx' due to IOException...
Deferring 'C:\ProgramData\Microsoft\Windows Defender\Support\MPDetection-20220126-183133.log' due to IOException...
Deferring 'C:\ProgramData\Microsoft\Windows Defender\Support\MPDeviceControl-20211016-164735.log' due to IOException...
Deferring 'C:\ProgramData\Microsoft\Windows Defender\Support\MPLog-10172021-040927.log' due to IOException...
Deferring 'C:\ProgramData\Microsoft\Windows Defender\Support\MpmpTracing-20220210-070038-00000003-ffffffff.bin' due to IOException...
Deferring 'C:\Windows\System32\winevt\logs\HardwareEvents.evtx' due to IOException...
Deferring 'C:\Windows\System32\winevt\logs\IntelAudioServiceLog.evtx' due to IOException...
Deferring 'C:\Windows\System32\winevt\logs\Internet Explorer.evtx' due to IOException...
.
.
.
.
Executing remaining modules...
Running 'EvtxECmd\EvtxECmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\EventLogs
Running 'JLECmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\FileFolderAccess -q
Running 'LECmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\FileFolderAccess -q
Running 'PECmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\ProgramExecution -q
Running 'RBCmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\FileDeletion -q
Running 'RECmd\RECmd.exe': -d C:\Users\Umais\Desktop\kape --bn BatchExamples\Kroll_Batch.reb --nl false --csv C:\Users\Umais\Desktop\4n6-2\BatchExamples
Running 'SBECmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\FileFolderAccess -q
Running 'SQLECmd\SQLECmd.exe': -d C:\Users\Umais\Desktop\kape --csv C:\Users\Umais\Desktop\4n6-2\SQLDatabases
Running 'SrumECmd.exe': -d C:\Users\Umais\Desktop\kape -k --csv C:\Users\Umais\Desktop\4n6-2\SystemActivity
Running 'SumECmd.exe': -d C:\Users\Umais\Desktop\kape\Windows\System32\LogFiles\SUM --csv C:\Users\Umais\Desktop\4n6-2\SUMDatabase

Executed 18 processors in 192.2738 seconds

Total execution time: 258.1812 seconds

Press any key to exit
```

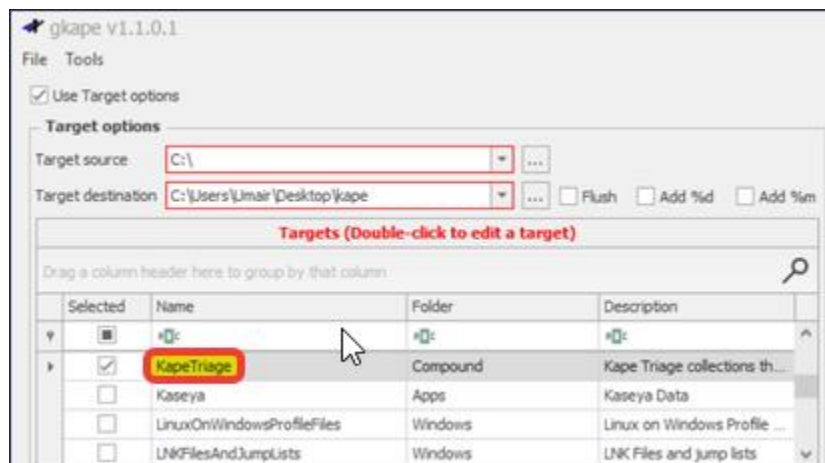
Notice that at the backend, KAPE is running the kape.exe in a command line. We can check out the files created by KAPE once it completes processing them. The below snapshot shows our Module destination. Notice how KAPE has processed the files according to different categories.

| Name                 | Date              | Type          | Size  | Tags |
|----------------------|-------------------|---------------|-------|------|
| EventLogs            | 1/26/2022 8:56 AM | File folder   |       |      |
| FileDeletion         | 1/26/2022 8:57 AM | File folder   |       |      |
| FileFolderAccess     | 1/26/2022 8:56 AM | File folder   |       |      |
| FileSystem           | 1/26/2022 8:55 AM | File folder   |       |      |
| ProgramExecution     | 1/26/2022 8:55 AM | File folder   |       |      |
| Registry             | 1/26/2022 8:57 AM | File folder   |       |      |
| SQLDatabases         | 1/26/2022 8:58 AM | File folder   |       |      |
| SUMDatabase          | 1/26/2022 8:58 AM | File folder   |       |      |
| SystemActivity       | 1/26/2022 8:58 AM | File folder   |       |      |
| 2022-01-26T035418... | 1/26/2022 8:58 AM | Text Document | 97 KB |      |

Let's collect triage data using the KAPETriage package, process it using !EZParser module, and answer the questions below. Then we can proceed to learn about the in the next task.

## Answer the questions below

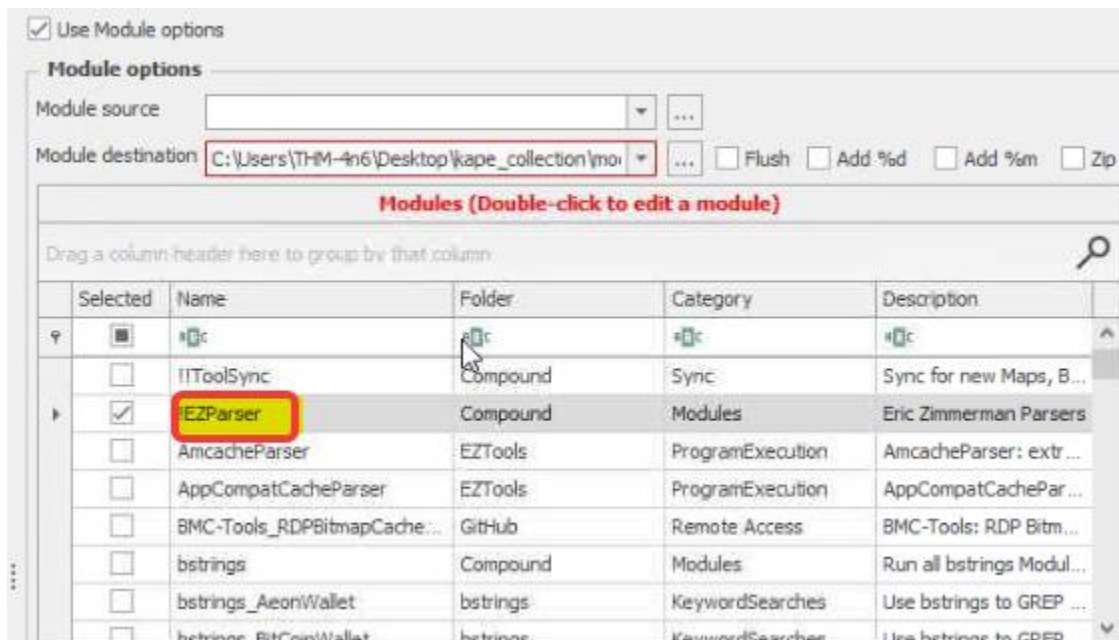
In the second to last screenshot above, what target have we selected for collection?



The selected target for collection is **KapeTriage**. This is shown in the gkape Target options window, where **KapeTriage** is checked under the Targets list. It is a compound target used to collect triage-related forensic artifacts.

Answer: KapeTriage

In the second to last screenshot above, what module have we selected for processing?



Answer: !EZParser

What option has to be checked to append date and time information to triage folder name?



To append date and time information to the triage folder name, the **Add %d** option must be checked. In KAPE, **%d** represents the timestamp, so enabling this option helps create a unique output folder for each collection.

Answer: %d

What option needs to be checked to add machine information to the triage folder name?



To add machine information to the triage folder name, the **Add %m** option must be checked. In KAPE, **%m** represents the machine name, so enabling this option adds the system name to the output folder.

Answer: %m

## Task 6 – KAPE CLI

Though we used the GUI in the previous task, KAPE is a command-line tool. Therefore, it is pertinent to know how to use KAPE through the command line to make full use of it.

For a list of all the different switches that can be used with KAPE, open an elevated PowerShell (Run As Administrator), go to the path where the KAPE binary is located, and type kape.exe. You will see something like this as an output.

### Target Collection Options

| Switch        | Description                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| --<br>tsource | Source drive or directory to collect files from. Examples: C, D:, F:, or a network path.                                                               |
| --target      | Target configuration to use for artifact collection.                                                                                                   |
| --tdest       | Destination directory where collected files will be saved. If --vhdx, --vhd, or --zip is used, the files will be stored inside that container instead. |
| --tlist       | Lists available targets. Use . to list all targets, or specify a Targets subdirectory.                                                                 |
| --tdetail     | Displays the details of a target configuration file.                                                                                                   |

| Switch   | Description                                                                                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --tflush | Deletes all existing files in the target destination directory before collection starts.                                                                                                |
| --tvars  | Provides key:value pairs for variable replacement in targets. Example: --tvars user:eric allows %user% in a target file to be replaced with eric. Multiple pairs are separated using ^. |
| --tdd    | Deduplicates collected files from --tsource, including Volume Shadow Copies when enabled, using SHA-1 hashes. The first matching file found is kept. Default: TRUE.                     |

## Module Processing Options

| Switch    | Description                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| --tsource | Directory containing the collected files to process. If targets are used and this option is blank, KAPE automatically uses --tdest.                   |
| --module  | Module configuration to use for processing collected artifacts.                                                                                       |
| --mdest   | Destination directory where module output files will be saved.                                                                                        |
| --mlist   | Lists available modules. Use . to list all modules, or specify a Modules subdirectory.                                                                |
| --mdetail | Displays the processor details for a module configuration file.                                                                                       |
| --mflush  | Deletes all existing files in the module destination directory before modules run.                                                                    |
| --mvars   | Provides key:value pairs for variable replacement in modules. Example: --mvars foo:bar replaces %foo% with bar. Multiple pairs are separated using ^. |
| --mef     | Sets the export format, such as csv, html, or json. This overrides the format configured in the module file.                                          |

## Collection, Simulation, and Volume Shadow Copies

| Switch | Description                                                                                                                                                                    |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --sim  | Simulates the collection process without copying files to --tdest. Default: FALSE.                                                                                             |
| --vss  | Processes all available Volume Shadow Copies on the source drive. Default: FALSE.                                                                                              |
| --hex  | Path to a file containing SHA-1 hashes to exclude. Files with hashes found in this file will not be copied.                                                                    |
| --ifw  | Controls KAPE behaviour when an FTK-related process is detected. By default, KAPE warns and exits. Set this option to true to ignore the warning and continue. Default: FALSE. |

## Container and Compression Options

| Switch | Description                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| --vhdx | Creates a VHDX container from the target destination. The value should be an identifier, not a filename. Cannot be used together with --vhd or --zip. |
| --vhd  | Creates a VHD container from the target destination. The value should be an identifier, not a filename. Cannot be used together with --vhdx or --zip. |
| --zip  | Creates a ZIP archive from the target destination. The value should be an identifier, not a filename. Cannot be used together with --vhd or --vhdx.   |
| --zv   | Compresses the created VHD or VHDX container into a ZIP file after creation. Default: TRUE.                                                           |
| --zm   | Compresses directories in --mdest into ZIP files. Default: FALSE.                                                                                     |
| --zpw  | Sets a password for ZIP files created through --zv, --zm, or --zip.                                                                                   |

## SFTP Transfer Options

| Switch  | Description                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------|
| --scs   | SFTP server hostname or IP address used for transferring compressed VHD or VHDX containers.                   |
| --scp   | SFTP server port. Default: 22.                                                                                |
| --scu   | SFTP username. Required when using --scs.                                                                     |
| --scpw  | SFTP password.                                                                                                |
| --scd   | Default destination directory on the SFTP server. The directory will be created if it does not already exist. |
| --scc   | Comment included with the SFTP transfer. By default, KAPE uses the name of the machine where it is running.   |
| --sftpc | Path to a configuration file containing SFTP server settings, such as ports and user accounts.                |
| --sftpu | Shows passwords in KAPE connection switches when using --sftpc. Default: TRUE.                                |
| --rlc   | Prevents KAPE from deleting the local copy after a successful transfer. Default: FALSE.                       |

## Amazon S3 and Cloud Storage Options

| Switch  | Description                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------|
| --s3p   | S3 provider name. Example: spAmazonS3 or spGoogleStorage. Default: spAmazonS3.                                       |
| --s3r   | S3 region name. Example: us-west-1 or ap-southeast-2.                                                                |
| --s3b   | S3 bucket name.                                                                                                      |
| --s3k   | S3 access key.                                                                                                       |
| --s3s   | S3 access secret.                                                                                                    |
| --s3st  | S3 session token.                                                                                                    |
| --s3kp  | S3 key prefix. This value is used as the beginning of the object key. Example: US1012/KapeData.                      |
| --s3o   | Object Storage Namespace used when the spOracle provider is selected.                                                |
| --s3c   | Comment included with the S3 transfer. By default, KAPE uses the name of the machine where it is running.            |
| --s3url | S3 presigned URL. This must use a PUT request rather than a GET request.                                             |
| --asu   | Azure Storage SAS URI.                                                                                               |
| --asc   | Comment included with the Azure Storage transfer. By default, KAPE uses the name of the machine where it is running. |

## General and Advanced Options

| Switch  | Description                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------|
| --debug | Displays debug information while KAPE is running.                                                                        |
| --trace | Displays detailed trace information while KAPE is running.                                                               |
| --gui   | Prevents KAPE from closing the command window when launched through gkape. Default: FALSE.                               |
| --ul    | When using _kape.cli, executes entries one at a time instead of in parallel. Default: FALSE.                             |
| --cu    | When using _kape.cli, deletes the _kape.cli file and both Target and Module directories when KAPE exits. Default: FALSE. |

| Switch  | Description                                                                                                                                                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --guids | Generates 10 GUIDs and exits. Useful when creating new targets or modules. Default: FALSE.                                                                                                                              |
| --sync  | Downloads the latest Targets and Modules before running KAPE. Default source: <a href="https://github.com/EricZimmerman/KapeFiles/archive/master.zip">https://github.com/EricZimmerman/KapeFiles/archive/master.zip</a> |

## KAPE Variables

| Variable | Meaning                                            |
|----------|----------------------------------------------------|
| %d       | Current timestamp in the format yyyyMMddTHH:mm:ss. |
| %s       | System drive letter.                               |
| %m       | Machine name.                                      |

## Example KAPE Commands

| Purpose                                                                    | Command                                                                                                     |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Collect Registry hives from drive L:                                       | kape.exe --tsource L: --target RegistryHives --tdest "c:\temp\RegistryOnly"                                 |
| Collect evidence of execution artifacts from drive H: with debug output    | kape.exe --tsource H --target EvidenceOfExecution --tdest "c:\temp\default" --debug                         |
| Collect Windows artifacts from a network location and store them in a VHDX | kape.exe --tsource \\server\directory\subdir --target Windows --tdest "c:\temp\default_%d" --vhdx LocalHost |
| Process collected data using the LECmd module                              | kape.exe --msource "c:\temp\default" --module LECmd --mdest "c:\temp\modulesOut" --trace --debug            |

We can see from the above screenshot that while collecting Targets, the switches tsource, target and tdest are required. Similarly, when processing files using Modules, module and mdest are required switches. The other switches are optional as per the requirements of the collection.

With this information, let's build a command to perform the same task we performed in the previous task. i.e., collect triage data using the KapeTriage Compound Target and process it

using the !EZParser Compound Module. Since we are not using the GUI version, we will start with typing:

```
kape.exe
```

To add a Target source, let's append --tsource and that Target path:

```
kape.exe --tsource C:
```

The --target flag will be used for selecting the Target the --tdest flag for the Target destination. For the sake of simplicity, we will set the Target destination to a directory named target on the Desktop. KAPE will create a new directory if it doesn't already exist. Our command line now looks like this:

```
kape.exe --tsource C: --target KapeTriage --tdest C:\Users\thm-4n6\Desktop\target
```

Running the above command will collect triage data defined in the KapeTriage Target and save it to the provided destination. However, it will not process it or perform any other activity on the data.

If we want to flush the Target destination, we can add --tflush to do that. For now, let's move on to adding the Module options. If we were using a Module source, we would have used a >--msource flag in a similar manner to the --tsource flag. But in this case, let's use the Target destination as the Module source. By doing this, we will not need to add it explicitly, and we can move on to adding the Module destination using the --mdest flag:

```
kape.exe --tsource C: --target KapeTriage --tdest C:\Users\thm-4n6\Desktop\target --mdest C:\Users\thm-4n6\Desktop\module
```

We have just used a directory named module for the Module destination.

To Process the Target destination using a Module, we need to provide the Module name using the --module flag. To process it using the !EZParser Module, we will append --module !EZParser, making our command look like this:

```
kape.exe --tsource C: --target KapeTriage --tdest C:\Users\thm-4n6\Desktop\target --mdest C:\Users\thm-4n6\Desktop\module --module !EZParser
```

Please note that we will need to run this command in an elevated shell (with Administrator privileges) for KAPE to collect the data.

We can modify the command as per our needs and the switches provided by KAPE. When we run this command, we will see a similar window as in the previous task. You can check out the files collected by KAPE Targets and Modules once it completes.

## Batch Mode:

KAPE can also be run in batch mode. What this means is that we can provide a list of commands for KAPE to run in a file named `_kape.cli`. Then we keep this file in the directory containing the KAPE binary. When `kape.exe` is executed as an administrator, it checks if there is `_kape.cli` file present in the directory. If so, it executes the commands mentioned in the file. This mode can be used if you need someone to run KAPE for you, you will keep all the commands in a single line, and all you need is for the person to right-click and run `kape.exe` as administrator. For example, if we have to perform the same task as we did earlier in this task using batch mode, we will have to create a `_kape.cli` file with the following content:

```
--tsource C: --target KapeTriage --tdest C:\Users\thm-4n6\Desktop\Target --mdest C:\Users\thm-4n6\Desktop\module --module !EZParser
```

When we run `kape.exe`, it will perform the same tasks as when we ran it through CLI above.

Let's answer the questions below and get ready to put our newly learned skills to use in the next task.

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe --tsource C: --target KapeTriage --tdest C:\Users\THM-4n6\Desktop\kape_collection\targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collection\modules\ --module !EZParser
KAPE version 1.1.0.1 Author: Eric Zimmerman (kape@kroll.com)
KAPE directory: C:\Users\THM-4n6\Desktop\KAPE
Command line: --tsource C: --target KapeTriage --tdest C:\Users\THM-4n6\Desktop\kape_collection\targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collection\modules\ --module !EZParser
System info: Machine name: THM-4n6, 64-bit: True, User: THM-4n6 OS: Windows10 (10.0.19044)
```

KAPE was then executed from the command line using the same options that were selected earlier in `gkape`. The command used **KapeTriage** as the target to collect triage artifacts from the C: drive, saved the collected files into the targets folder, and used **EZParser** as the module to process the collected artifacts.

Command used:

```
.\kape.exe --tsource C: --target KapeTriage --tdest C:\Users\THM-4n6\Desktop\kape_collection\targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collection\modules\ --module !EZParser
```

The output confirms that KAPE started successfully and detected the system information, including the machine name **THM-4N6**, the logged-in user **THM-4n6**, and the operating system **Windows 10**.

## Answer the questions below

Run the command `kape.exe` in an elevated shell. Take a look at the different switches and variables. What variable adds the collection timestamp to the target destination?

```
Administrator: Windows PowerShell
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe | Select-String "timestamp"

Variables: %d = Timestamp (yyyyMMddTHH:mm:ss)
```

The variable used to add the collection timestamp to the target destination is `%d`. KAPE shows this variable as the timestamp format **yyyyMMddTHH:mm:ss**, which can be appended to the output folder name to make each collection folder unique.

Command used:

```
.\kape.exe | Select-String "timestamp"
```

Answer: %d

What variable adds the machine information to the target destination?

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe | Select-String "machine name"

% m = Machine name
```

The variable used to add machine information to the target destination is `%m`. KAPE shows this variable as the **Machine name**, which can be added to the output folder name to identify which system the collection came from.

Command used:

```
.\kape.exe | Select-String "machine name"
```

Answer: %m

Which switch can be used to show debug information during processing?

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe | Select-String "debug"
debug          Show debug information during processing
kape.exe --tsource H --target EvidenceOfExecution --tdest "c:\temp\default" --debug
kape.exe --msource "c:\temp\default" --module LECmd --mdest "c:\temp\modulesOut" --trace --debug
```

The switch used to show debug information during KAPE processing is **--debug**. This can be added to a KAPE command when more detailed output is needed during collection or module processing.

Command used:

```
.\kape.exe | Select-String "debug"
```

Answer: debug

Which switch is used to list all targets available?

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe | Select-String "target"
tsource      Target source drive to copy files from (C, D:, or F:\ for example)
target       Target configuration to use
tlist        List available Targets. Use . for Targets directory or name of subdirectory under Targets.
tdestall     Dump Target file details
tvars        Provide a list of key:value pairs to be used for variable replacement in Targets. Ex: --tvars user:eric would allow for using %user% in a Target which is replaced with eric at runtime. Multiple pairs should be separated by ^
msource      Directory containing files to process. If using Targets and this is left blank, it will be set to --tdest automatically
cu           When using _kape.cli, if true, KAPE will delete _kape.cli and both Target/Module directories upon exiting. Default is FALSE
guids        KAPE will generate 10 GUIDs and exit. Useful when creating new Targets/Modules. Default is FALSE
sync         If true, KAPE will download the latest Targets and Modules from specified URL prior to running. Default is https://github.com/EricZimmerman/Kapefiles/archive/master.zip
Examples: kape.exe --tsource L: --target RegistryHives --tdest "c:\temp\RegistryOnly"
kape.exe --tsource H --target EvidenceOfExecution --tdest "c:\temp\default" --debug
kape.exe --tsource \\server\directory\subdir --target windows --tdest "c:\temp\default_xd" --vhd LocalHost
```

The switch used to list all available KAPE targets is **--tlist**. This option shows the target configurations that can be used for artifact collection.

Command used:

```
.\kape.exe | Select-String "target"
```

Answer: tlist

Which flag, when used with batch mode, will delete the `_kape.cli`, targets and modules files after the execution is complete?

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe | Select-String "delete"

tflush      Delete all files in 'tdest' prior to collection
mflush      Delete all files in 'mdest' prior to running Modules
cu          When using _kape.cli, if true, KAPE will delete _kape.cli and both Target/Module directories upon exiting. Default is FALSE
rlc         If true, local copy of transferred files will NOT be deleted after upload. Default is FALSE
```

The flag used with batch mode to delete `_kape.cli`, the Target directory, and the Module directory after execution is `--cu`. In the KAPE help output, this option is shown as `cu`, and it applies when KAPE is run using `_kape.cli` batch mode.

Command used:

```
.\kape.exe | Select-String "delete"
```

Answer: cu

## Task 7 – Hands-on Challenge

So, now that we have learned how to use KAPE let's put it into practice. For this task, you will need to utilize your skills gained in this room and the previous [Windows Forensics 1](#) and [Windows Forensics 2](#) rooms.

Organization X has an Acceptable Use Policy for their Portable Devices, including Laptops. This policy forbids users from connecting removable or Network drives, installing software from unknown locations, and connecting to unknown networks. It looks like one of the users has violated this policy. Can you help Organization X find out if the user violated the Acceptable Use Policy on their device? The user's machine is attached to the room as a VM.

Navigate to the KAPE directory placed on the Desktop in the attached VM. Run KAPE with your desired Target and Module options, and answer the following questions.

**Hint:** You can use EZviewer placed in the EZtools folder on Desktop to open CSV files.

## Answer the questions below

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe --source C: --target KapeTriage --tdest C:\Users\THM-4n6\Desktop\kape_collection\Targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collection\Modules\ --module IEZParser
KAPE version 1.1.0.1 Author: Eric Zimmerman (kape@roll.com)

KAPE directory: C:\Users\THM-4n6\Desktop\KAPE
Command line: --source C: --target KapeTriage --tdest C:\Users\THM-4n6\Desktop\kape_collection\Targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collection\Modules\ --module IEZParser

System info: Machine name: THM-4n6, 64-bit: True, User: THM-4n6 OS: Windows10 (10.0.19044)
```

### KAPE Collection and Processing

I first ran KAPE using the **KapeTriage** target and the **IEZParser** module to collect and process common forensic artifacts from the lab machine. After the collection finished, I used both the parsed module output and the raw target output to answer the questions. If the parsed output did not contain the needed evidence, I reviewed the original files collected under the target output folder.

Command used:

```
.\kape.exe --tsource C: --target KapeTriage --tdest C:\Users\THM-4n6\Desktop\kape_collection\Targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collection\Modules\ --module !EZParser
```

Two USB Mass Storage devices were attached to this Lab Machine. One had a Serial Number 0123456789ABCDE. What is the Serial Number of the other USB Device?

```
PS C:\Users\THM-4n6\Desktop\KAPE> .\kape.exe --source C: --target USBDevicesLogs --tdest C:\Users\THM-4n6\Desktop\kape_collections\usb_targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collections\module_targets\ --module IEZParser
KAPE version 1.1.0.1 Author: Eric Zimmerman (kape@roll.com)

KAPE directory: C:\Users\THM-4n6\Desktop\KAPE
Command line: --source C: --target USBDevicesLogs --tdest C:\Users\THM-4n6\Desktop\kape_collections\usb_targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collections\module_targets\ --module IEZParser

System info: Machine name: THM-4n6, 64-bit: True, User: THM-4n6 OS: Windows10 (10.0.19044)

Using Target operations
  Creating target destination directory 'C:\Users\THM-4n6\Desktop\kape_collections\usb_targets'
Found 3 targets. Expanding targets to file list...
Found 1 file in 0.090 seconds. Beginning copy...

Copied 1 out of 1 files in 1.6552 seconds. See "Copylog.csv" in 'C:\Users\THM-4n6\Desktop\kape_collections\usb_targets' for copy details

Using Module operations
  Setting --source to 'C:\Users\THM-4n6\Desktop\kape_collections\usb_targets' since --source was not provided
  Creating module destination directory 'C:\Users\THM-4n6\Desktop\kape_collections\module_targets'
```

```
PS C:\Users\THM-4n6\Desktop\KAPE> Get-Content C:\Users\THM-4n6\Desktop\kape_collections\usb_targets\C\Windows\inf\setupapi.dev.log | Select-String "USBSTOR"

! set:  USBSTOR\DISK&VEN_KINGSTON&PROD_DATATRAVELER_2.0&REV_PMAP\1C6F654E59A3B0C1790366AE80 -> Configured [disk.inf:GenDisk,disk_install.NT] and non-present (ConfigFlags = 0x00000000).
! set:  USBSTOR\DISK&VEN_USB3.0&REV_EXTERNAL_DEVICE&REV_SDMA\0123456789ABCDE80 -> Configured [disk.inf:GenDisk,disk_install.NT] and non-present (ConfigFlags = 0x00000000).
! set:  USB\VID_0930&PID_654E\1C6F654E59A3B0C1790366AE80 -> Configured [usbstor.inf:USB\Class_08&Subclass_06&Prot_50,USBSTOR_BULK.NT] and non-present (ConfigFlags = 0x00000000).
! set:  USB\VID_2537&PID_1068\0123456789ABCDE -> Configured [usbstor.inf:USB\Class_08&Subclass_06&Prot_50,USBSTOR_BULK.NT] and non-present (ConfigFlags = 0x00000000).
! set:  USBSTOR\DISK&VEN_KINGSTON&PROD_DATATRAVELER_2.0&REV_PMAP\1C6F654E59A3B0C1790366AE80 -> Configured [disk.inf:GenDisk,disk_install.NT] and non-present (ConfigFlags = 0x00000000).
```

| Key name              | # values | # subkeys | Last write timestamp |
|-----------------------|----------|-----------|----------------------|
| ACPI                  | 0        | 21        | 2021-11-24 18:40:54  |
| ACPI_HAL              | 0        | 1         | 2021-11-24 18:40:40  |
| BTM                   | 0        | 3         | 2021-11-24 18:40:47  |
| DISPLAY               | 0        | 1         | 2021-11-24 18:40:47  |
| CONSOLE               | 0        | 1         | 2021-11-24 18:40:46  |
| HID                   | 0        | 2         | 2021-11-24 18:40:44  |
| WTRFEE                | 0        | 1         | 2021-11-24 18:40:40  |
| SR                    | 0        | 1         | 2021-11-24 18:40:46  |
| PCI                   | 0        | 20        | 2021-11-24 18:40:03  |
| PCIDE                 | 0        | 1         | 2021-11-24 18:40:40  |
| PCIDE                 | 0        | 18        | 2021-09-17 17:28:03  |
| ACPI                  | 0        | 3         | 2021-11-24 18:40:46  |
| STORAGE               | 0        | 1         | 2021-11-24 18:40:44  |
| SMI                   | 0        | 9         | 2021-11-24 18:40:47  |
| TRANSPORT_BULK        | 0        | 1         | 2021-11-24 18:40:46  |
| TS_USB_HUB_Enumerator | 0        | 1         | 2021-11-24 18:40:46  |
| USB                   | 0        | 10        | 2021-11-24 18:40:46  |

| Timestamp           | Manufacturer | Title                 | Version  | Disk ID                                  | Serial Number     | Device Name                          | Installed           | First Installed     | Last Connected      | Last Removed |
|---------------------|--------------|-----------------------|----------|------------------------------------------|-------------------|--------------------------------------|---------------------|---------------------|---------------------|--------------|
| 2021-11-24 18:25:15 | Ver_Storage  | Prod_DataTraveler_2.0 | Rev_PMAP | {6295931F-4862-11e1-a3c1-3-00ca76da7110} | 0123456789ABCDE80 | Kingston DataTraveler 2.0 USB Device | 2021-11-24 18:25:15 | 2021-11-24 18:25:15 | 2021-11-24 18:40:06 | --           |
| 2021-11-24 18:27:02 | Ver_USB3.0   | Prod_External_Device  | Rev_SDMA | {F23646-4616-11e1-a3c1-2-00ca76da7110}   | 0123456789ABCDE   | USB 3.0 External Device USB Device   | 2021-11-24 18:27:02 | 2021-11-24 18:27:02 | 2021-11-24 18:27:02 | --           |

After running the main **KapeTriage** collection, I could not find the specific USB artifact needed for this question. The required file was **setupapi.dev.log**, which contains USB

device installation and configuration records. Since this file was not present in the main KapeTriage output, I ran a focused KAPE collection using the **USBDevicesLogs** target.

Command used:

```
.\kape.exe --tsource C: --target USBDevicesLogs --tdest C:\Users\THM-4n6\Desktop\kape_collections\usb_targets\ --mdest C:\Users\THM-4n6\Desktop\kape_collections\module_targets\ --module !EZParser
```

I then searched the collected **setupapi.dev.log** file for USBSTOR entries.

Command used:

```
Get-Content C:\Users\THM-4n6\Desktop\kape_collections\usb_targets\C\Windows\inf\setupapi.dev.log |  
Select-String "USBSTOR"
```

The output showed two USB Mass Storage serial numbers. One was the known serial number **0123456789ABCDE**, and the other USB device serial number was:

Answer: 1C6F654E59A3B0C179D366AE

7zip, Google Chrome and Mozilla Firefox were installed from a Network drive location on the Lab Machine. What was the drive letter and path of the directory from where these software were installed?

EZViewer v1.0.0.0 - 20260711155654\_RecentApps\_C\_Users\_THM-4n6\_NTUSER.DAT.csv

|    | A                  | B                                                                            | C             | D         | E          | F                | G        | H                   | I |
|----|--------------------|------------------------------------------------------------------------------|---------------|-----------|------------|------------------|----------|---------------------|---|
| 1  | KeyName            | BatchKey                                                                     | AppId         | BatchValu | AppPath    | LastAccessed     | LaunchCo | RecentDocs          |   |
| 2  | {084A7F26-ROOT\SOI | Z:\setups\Firefox                                                            | Installer.exe | Multiple  |            | 11/25/2021 3:27  | 1        |                     |   |
| 3  | {207E2FD2-ROOT\SOI | E7CF176E110C211B                                                             |               | Multiple  | C:\Progra  | 11/25/2021 3:44  | 1        |                     |   |
| 4  | {273DE51E-ROOT\SOI | {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\WindowsPowerShell\v1.0\powershell.exe |               | Multiple  | C:\Windo   | 11/25/2021 3:37  | 2        |                     |   |
| 5  | {2CD6F947-ROOT\SOI | Microsoft.Windows.RemoteDesktop                                              |               | Multiple  | C:\Windo   | 11/25/2021 3:59  | 1        |                     |   |
| 6  | {48ED5797-ROOT\SOI | D:\setup64.exe                                                               |               | Multiple  |            | 11/25/2021 3:21  | 1        |                     |   |
| 7  | {56C40D17-ROOT\SOI | Z:\setups\python-3.8.2.exe                                                   |               | Multiple  |            | 11/25/2021 3:32  | 1        |                     |   |
| 8  | {6E5F6115-ROOT\SOI | C:\Users\THM-4n6\Desktop\KAPE\KAPE\gkape.exe                                 |               | Multiple  |            | 11/25/2021 3:44  | 2        |                     |   |
| 9  | {75D4CE35-ROOT\SOI | C:\Users\THM-4n6\Downloads\Windows10Upgrade9252.exe                          |               | Multiple  |            | 11/24/2021 15:24 | 1        |                     |   |
| 10 | {7EEDBC81-ROOT\SOI | Z:\setups\7z1900-x64.exe                                                     |               | Multiple  |            | 11/25/2021 3:26  | 1        |                     |   |
| 11 | {9A9AB561-ROOT\SOI | C:\Users\THM-4n6\Downloads\MicrosoftEdgeSetup.exe                            |               | Multiple  |            | 11/25/2021 4:12  | 1        |                     |   |
| 12 | {AF13CF4C-ROOT\SOI | Microsoft.Windows.Explorer                                                   |               | Multiple  | ::{52205FC | 11/24/2021 15:24 | 16       |                     |   |
| 13 | {C2CDA02-ROOT\SOI  | OperaSoftware.OperaWebBrowser.1637811529                                     |               | Multiple  | C:\Users\  | 11/25/2021 4:10  | 1        |                     |   |
| 14 | {E08EEF4E-ROOT\SOI | Z:\setups\OperaSetup.exe                                                     |               | Multiple  |            | 11/25/2021 3:33  | 1        |                     |   |
| 15 | {EB50F050-ROOT\SOI | Z:\setups\ChromeSetup.exe                                                    |               | Multiple  |            | 11/25/2021 3:33  | 1        |                     |   |
| 16 | {FB7FFC3C-ROOT\SOI | {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\NOTEPAD.EXE                           |               | Multiple  | C:\Windo   | 11/25/2021 3:42  | 9        | ChangeLog/C:\Users\ |   |
| 17 | {FC5827CC-ROOT\SOI | {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\SystemPropertiesComputerName.exe      |               | Multiple  |            | 11/25/2021 4:01  | 1        |                     |   |

C:\Users\THM-4n6\Desktop\kape\_triage\modules\Registry\20260711155654\20260711155654\_RecentApps\_C\_Users\_THM-4n6\_NTUSER.DAT.csv

I found the installation source path in the parsed Registry output. In EZViewer, I opened the RecentApps CSV parsed from the user's NTUSER.DAT hive. The entries showed that Firefox Installer.exe, 7z1900-x64.exe, and ChromeSetup.exe were all executed from the same network drive location.

Parsed file reviewed:

C:\Users\THM-4n6\Desktop\kape\_triage\modules\Registry\20260711155654\20260711155654\_RecentApps\_C\_Users\_THM-4n6\_NTUSER.DAT.csv

The common source path was:

Answer: Z:\Setups

What is the execution date and time of CHROMESETUP.EXE in MM/DD/YYYY HH:MM?

| EZViewer v1.0.0.0 - 20260711155654_RecentApps_C_Users_THM-4n6_NTUSER.DAT.csv |            |           |                                                                              |           |            |                  |          |            |           |
|------------------------------------------------------------------------------|------------|-----------|------------------------------------------------------------------------------|-----------|------------|------------------|----------|------------|-----------|
| File Tools Help                                                              |            |           |                                                                              |           |            |                  |          |            |           |
|                                                                              | A          | B         | C                                                                            | D         | E          | F                | G        | H          | I         |
| 1                                                                            | KeyName    | BatchKeyf | AppId                                                                        | BatchValu | AppPath    | LastAccessed     | LaunchCo | RecentDocs |           |
| 2                                                                            | {084A7F26} | ROOT\SOI  | Z:\setups\Firefox Installer.exe                                              | Multiple  |            | 11/25/2021 3:27  | 1        |            |           |
| 3                                                                            | {207E2FD2} | ROOT\SOI  | E7CF176E110C211B                                                             | Multiple  | C:\Progra  | 11/25/2021 3:44  | 1        |            |           |
| 4                                                                            | {273DE518} | ROOT\SOI  | {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\WindowsPowerShell\v1.0\powershell.exe | Multiple  | C:\Windo   | 11/25/2021 3:37  | 2        |            |           |
| 5                                                                            | {2CD6F94}  | ROOT\SOI  | Microsoft.Windows.RemoteDesktop                                              | Multiple  | C:\Windo   | 11/25/2021 3:59  | 1        |            |           |
| 6                                                                            | {48ED5797} | ROOT\SOI  | D:\setup64.exe                                                               | Multiple  |            | 11/25/2021 3:21  | 1        |            |           |
| 7                                                                            | {56C40D12} | ROOT\SOI  | Z:\setups\python-3.8.2.exe                                                   | Multiple  |            | 11/25/2021 3:32  | 1        |            |           |
| 8                                                                            | {6E5F6115} | ROOT\SOI  | C:\Users\THM-4n6\Desktop\KAPE\KAPE\kape.exe                                  | Multiple  |            | 11/25/2021 3:44  | 2        |            |           |
| 9                                                                            | {75D4CE35} | ROOT\SOI  | C:\Users\THM-4n6\Downloads\Windows10Upgrade9252.exe                          | Multiple  |            | 11/24/2021 15:24 | 1        |            |           |
| 10                                                                           | {7EEDBC8}  | ROOT\SOI  | Z:\setups\7z1900-x64.exe                                                     | Multiple  |            | 11/25/2021 3:26  | 1        |            |           |
| 11                                                                           | {9A9AB56}  | ROOT\SOI  | C:\Users\THM-4n6\Downloads\MicrosoftEdgeSetup.exe                            | Multiple  |            | 11/25/2021 4:12  | 1        |            |           |
| 12                                                                           | {AF13CF4}  | ROOT\SOI  | Microsoft.Windows.Explorer                                                   | Multiple  | ::{52205FC | 11/24/2021 15:24 | 16       |            |           |
| 13                                                                           | {C2CDA02}  | ROOT\SOI  | OperaSoftware.OperaWebBrowser.1637811529                                     | Multiple  | C:\Users\  | 11/25/2021 4:10  | 1        |            |           |
| 14                                                                           | {E08EEF4E} | ROOT\SOI  | Z:\setups\OperaSetup.exe                                                     | Multiple  |            | 11/25/2021 3:33  | 1        |            |           |
| 15                                                                           | {EB50F050} | ROOT\SOI  | Z:\setups\ChromeSetup.exe                                                    | Multiple  |            | 11/25/2021 3:33  | 1        |            |           |
| 16                                                                           | {FB7FFC5}  | ROOT\SOI  | {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\NOTEPAD.EXE                           | Multiple  | C:\Windo   | 11/25/2021 3:42  | 9        | ChangeLog\ | C:\Users\ |
| 17                                                                           | {FC5827C}  | ROOT\SOI  | {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\SystemPropertiesComputerName.exe      | Multiple  |            | 11/25/2021 4:01  | 1        |            |           |

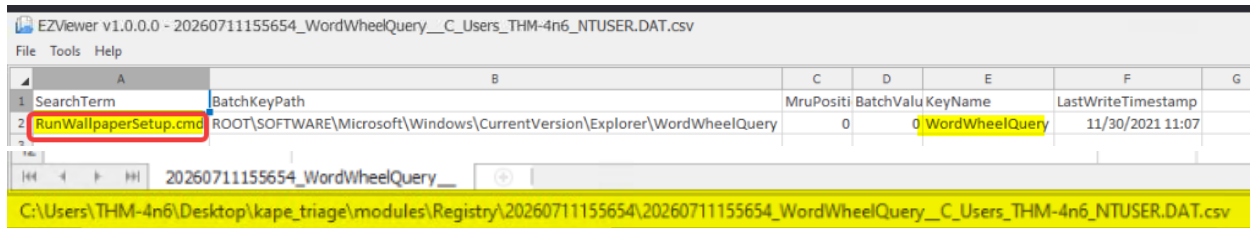
I found the execution time in the parsed **RecentApps** Registry output from the user's **NTUSER.DAT** hive. In EZViewer, the entry for **Z:\setups\ChromeSetup.exe** showed the **LastAccessed** time as **11/25/2021 03:33**.

Parsed file reviewed:

C:\Users\THM-4n6\Desktop\kape\_triage\modules\Registry\20260711155654\20260711155654\_RecentApps\_C\_Users\_THM-4n6\_NTUSER.DAT.csv

Answer: Answer: **11/25/2021 03:33**

What search query was run on the system?



The screenshot shows the EZViewer application displaying a CSV file named '20260711155654\_WordWheelQuery\_\_C\_Users\_THM-4n6\_NTUSER.DAT.csv'. The file contains a single row of data with the following values:

| SearchTerm            | BatchKeyPath                                                           | MruPositi | BatchValu | KeyName        | LastWriteTimestamp |
|-----------------------|------------------------------------------------------------------------|-----------|-----------|----------------|--------------------|
| RunWallpaperSetup.cmd | ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\WordWheelQuery | 0         | 0         | WordWheelQuery | 11/30/2021 11:07   |

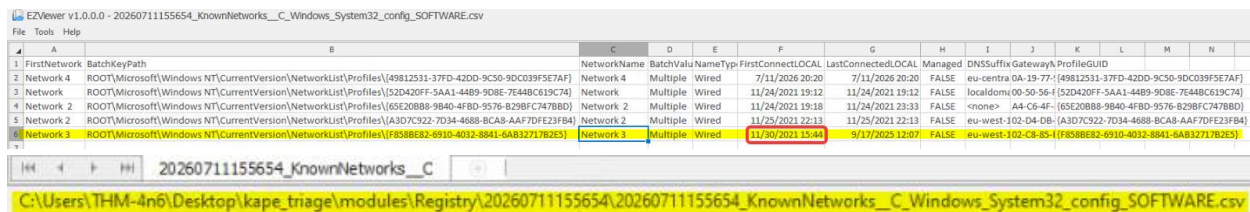
I found the search query in the parsed **WordWheelQuery** Registry output from the user's **NTUSER.DAT** hive. This artifact stores Windows Explorer search terms. In EZViewer, the **SearchTerm** value showed that the query run on the system was **RunWallpaperSetup.cmd**.

Parsed file reviewed:

C:\Users\THM-4n6\Desktop\kape\_triage\modules\Registry\20260711155654\20260711155654\_WordWheelQuery\_\_C\_Users\_THM-4n6\_NTUSER.DAT.csv

Answer: RunWallpaperSetup.cmd

When was the network named Network 3 First connected to?



The screenshot shows the EZViewer application displaying a CSV file named '20260711155654\_KnownNetworks\_\_C\_Windows\_System32\_config\_SOFTWARE.csv'. The file contains a table with network connection data. The row for Network 3 is highlighted, showing the FirstConnectedLOCAL value as 11/30/2021 15:44.

| NetworkName | BatchValu | NameType | FirstConnectLOCAL | LastConnectedLOCAL | Managed | DNSSuffix | Gateway                                           | ProfileGUID |
|-------------|-----------|----------|-------------------|--------------------|---------|-----------|---------------------------------------------------|-------------|
| Network 4   | Multiple  | Wired    | 7/11/2026 20:20   | 7/11/2026 20:20    | FALSE   | eu-centra | 0A-19-77-{:49812531-37FD-42DD-9C50-9DC039F5E7AF}  |             |
| Network 2   | Multiple  | Wired    | 11/24/2021 19:12  | 11/24/2021 19:12   | FALSE   | localdom  | 00-50-56-{:52D420FF-5AA1-44B9-90BE-764BC615C7A}   |             |
| Network 2   | Multiple  | Wired    | 11/24/2021 19:18  | 11/24/2021 23:33   | FALSE   | <none>    | A4-C6-4F-{:65E208B8-9B40-4FBD-9576-8298FC7478BD}  |             |
| Network 3   | Multiple  | Wired    | 11/30/2021 15:44  | 9/17/2025 12:07    | FALSE   | eu-west   | 102-04-DB-{:A3D7C922-7D34-4668-BCA8-AAF7DFE23FB4} |             |

I found this in the parsed **KnownNetworks** Registry output from the **SOFTWARE** hive. In EZViewer, the row for **Network 3** showed the **FirstConnectedLOCAL** value as **11/30/2021 15:44**.

Parsed file reviewed:

C:\Users\THM-4n6\Desktop\kape\_triage\modules\Registry\20260711155654\20260711155654\_KnownNetworks\_\_C\_Windows\_System32\_config\_SOFTWARE.csv

Answer: 11/30/2021 15:44

KAPE was copied from a removable drive. Can you find out what was the drive letter of the drive where KAPE was copied from?

EZViewer v1.0.0.0 - 20260711205536\_AutomaticDestinations.csv

| A    | B         | C      | D      | E     | F     | G                                                      | H    | I    | J     | K        | L    | M    | N   | O                                                                                                   | P     | Q     | R                                   | S    | T    | U    | V    |
|------|-----------|--------|--------|-------|-------|--------------------------------------------------------|------|------|-------|----------|------|------|-----|-----------------------------------------------------------------------------------------------------|-------|-------|-------------------------------------|------|------|------|------|
| File | Tools     | Help   |        |       |       |                                                        |      |      |       |          |      |      |     |                                                                                                     |       |       |                                     |      |      |      |      |
| 1    | Source    | Source | Source | AppId | AppId | Dest                                                   | Last | Last | Entry | Creation | Last | Home | Mac | Path                                                                                                | Inter | Pre   | File                                | File | File | File | File |
| 2    | C:\Users\ |        |        |       |       | 375015ed Quick As                                      | 4    | 6    | 0     |          |      |      |     | 02:53a79 C:\ProgramData\Sysmon.zip                                                                  | 1     | FALSE | ee05a054-ee05a054-7eb87548-7eb87548 |      |      |      |      |
| 3    | C:\Users\ |        |        |       |       | 375015ed Quick As                                      | 4    | 6    | 1     | 3        |      |      |     | 02:5b7c7 C:\Program Files\Amazon\IzConfigService\Settings\WallpaperSettings.xml                     | 1     | FALSE | ee05a054-ee05a054-7eb87548-7eb87548 |      |      |      |      |
| 4    | C:\Users\ |        |        |       |       | 7e4d480 Control Pa                                     | 4    | 1    | 0     | 1        |      |      |     | 28E0968-A00A-4407-9371-8E8064C3883\01-3B80C0E4-0293-4F75-8A90-C8058647EE1E ==> Control Panel\System | 1     | FALSE | ee05a054-ee05a054-7eb87548-7eb87548 |      |      |      |      |
| 5    | C:\Users\ |        |        |       |       | 9b0cd09c Notepad f                                     | 4    | 4    | 0     | 4        |      |      |     | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE\Changing.txt                                                 | 1     | FALSE | 076c1206-076c1206-7eb87548-7eb87548 |      |      |      |      |
| 6    | C:\Users\ |        |        |       |       | 9b0cd09c Notepad f                                     | 4    | 4    | 1     | 3        |      |      |     | 02:5b7c7 C:\Program Files\Amazon\IzConfigService\Settings\WallpaperSettings.xml                     | 1     | FALSE | ee05a054-ee05a054-7eb87548-7eb87548 |      |      |      |      |
| 7    | C:\Users\ |        |        |       |       | 9b0cd09c Notepad f                                     | 4    | 4    | 2     | 2        |      |      |     | 00:1a70c C:\Users\THM-4n6\Desktop\KAPE\IzConfigService\Settings\WallpaperSettings.xml               | 1     | FALSE | 076c1206-076c1206-7eb87548-7eb87548 |      |      |      |      |
| 8    | C:\Users\ |        |        |       |       | 9b0cd09c Notepad f                                     | 4    | 4    | 3     | 1        |      |      |     | 00:1a70c C:\Users\THM-4n6\Desktop\KAPE\IzConfigService\Settings\WallpaperSettings.xml               | 1     | FALSE | 076c1206-076c1206-7eb87548-7eb87548 |      |      |      |      |
| 9    | C:\Users\ |        |        |       |       | d4702b1a Microsoft                                     | 4    | 1    | 0     | 1        |      |      |     | 00:1a70c C:\Users\THM-4n6\Desktop\KAPE\IzConfigService\Settings\WallpaperSettings.xml               | 1     | FALSE | 076c1206-076c1206-7eb87548-7eb87548 |      |      |      |      |
| 10   | C:\Users\ |        |        |       |       | 02:53a79 C:\ProgramData\Sysmon                         | 4    | 31   | 0     | 1F       |      |      |     | 02:53a79 C:\ProgramData\Sysmon                                                                      | 1     | FALSE | ee05a054-ee05a054-7eb87548-7eb87548 |      |      |      |      |
| 11   | C:\Users\ |        |        |       |       | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE                 | 4    | 31   | 1     | 1E       |      |      |     | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE                                                              | 1     | FALSE | 076c1206-076c1206-7eb87548-7eb87548 |      |      |      |      |
| 12   | C:\Users\ |        |        |       |       | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE                 | 4    | 31   | 2     | 1        |      |      |     | 00:1a70c C:\Users\THM-4n6\Desktop\KAPE\IzConfigService\Settings\WallpaperSettings.xml               | 10    | TRUE  | 754efb42-754efb42-7eb87548-7eb87548 |      |      |      |      |
| 13   | C:\Users\ |        |        |       |       | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE                 | 4    | 31   | 3     | 12       |      |      |     | 00:1a70c C:\Users\THM-4n6\Desktop\KAPE\IzConfigService\Settings\WallpaperSettings.xml               | 2     | FALSE | 754efb42-754efb42-7eb87548-7eb87548 |      |      |      |      |
| 14   | C:\Users\ |        |        |       |       | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE                 | 4    | 31   | 4     | 2        |      |      |     | 00:1a70c C:\Users\THM-4n6\Desktop\KAPE\IzConfigService\Settings\WallpaperSettings.xml               | 4     | TRUE  | 754efb42-754efb42-7eb87548-7eb87548 |      |      |      |      |
| 15   | C:\Users\ |        |        |       |       | 02:78a15 C:\Users\THM-4n6\Desktop\KAPE                 | 4    | 31   | 5     | 1D       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 16   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 6     | 1C       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 17   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 7     | 1B       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 18   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 8     | 1A       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 19   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 9     | 19       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 20   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 10    | 12       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 21   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 11    | 17       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 22   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 12    | 16       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 23   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 13    | 1F       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 24   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 14    | 12       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 25   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 15    | 14       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 26   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 16    | 11       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 27   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 17    | 10       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 28   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 18    | 1E       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 29   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 19    | 1D       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 30   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 20    | 1A       |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |
| 31   | C:\Users\ |        |        |       |       | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer | 4    | 31   | 21    | 7        |      |      |     | 02:29032 C:\Users\THM-4n6\Desktop\EZtools\SOBEexplorer                                              | 1     | FALSE | a7865e05-a7865e05-7eb87548-7eb87548 |      |      |      |      |

20260711205536\_AutomaticDestina

C:\Users\THM-4n6\Desktop\kape\_triage\modules\FileFolderAccess\20260711205536\_AutomaticDestinations.csv

I found this in the parsed **FileFolderAccess** output. In EZViewer, I opened the **AutomaticDestinations.csv** file and reviewed the recorded folder paths. The entry **E:\KAPE** showed that KAPE had been accessed from drive **E:**, which indicates the removable drive letter.

Parsed file reviewed:

**C:\Users\THM-4n6\Desktop\kape\_triage\modules\FileFolderAccess\20260711205536\_AutomaticDestinations.csv**

Answer: E:

Disclaimer

This writeup was created for educational and portfolio purposes only. All testing was performed in the relevant lab environment.

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